CHAPTER

2 Probability and Inference

Historically, logic and logical reasoning have played a central role in artificial intelligence. However, pure Boolean logic is often insufficent to capture the complexity of interacting with the real world. Instead, it is often important to reason about quantities and events that are *uncertain*: to make predictions, and to assess our degree of uncertainty about those predictions.

In practice, random effects may arise from a number of potential sources. Often, randomness is a convenient substitute for unmodelled complexity; for example, suppose that we are planning a trip to the airport, and would like to predict how long it will take. If we knew the exact schedule and plans of every driver in our city, perhaps we could determine the amount of traffic at each point along our route, at the time we would pass by, and there would be no real randomness at all. However, not only do we lack this knoweldge, but even if we possessed it the resulting model would have overwhelming complexity. Instead, it is far simpler to treat the system as being an accumulation of random effects.

Probability gives us a useful framework for quantifying our uncertainty about the world. In particular, probability theory gives us a natural way to describe our assumptions, with an ability to express our degree of uncertainty. Moreover, it gives rules for how to reason about systems of uncertain quantities, including how to combine information from multiple sources, and how our uncertainty is changed when some parts of the system are observed.

2.1 Probability, Events, Random Variables

We begin with some basic definitions that will allow us to talk about random events and random variables, e.g., quantities whose values are uncertain.

Axioms of Probability

In probability theory, we define the *event space* S to be the set of all possible outcomes, or (loosely speaking) all possible worlds that could happen. Then, a *random event* is simply a subset of those outcomes, $A \subseteq S$. Typically, this subset corresponds to some concept of interest, such as "the set of all outcomes where my coin toss comes up heads". A random event is thus a boolean statement which may be either true or false.

Then, a *probability measure* assigns a probability to all possible events. More formally, a probability measure or probability mass function $\Pr[\cdot]$ is a function over possible events A, which obeys the following properties:

1. $\forall A, \qquad 0 \leq \Pr[A] \leq 1$
2. $\Pr[S] = 1$
3. $\Pr[\varnothing] = 0$
4. $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$

The first three of these are quite straightforward; the probability of any event A is between zero and one, with the probability of an event that is deterministically true (true in all possible worlds) having the maximum possible value, one; and the probability of an event that is deterministically false (false in all possible worlds) having the minimum value, zero.

The last property tells us how to reason about compositions of events. Viewing the probability as the fraction of possible worlds, it says that to compute the fraction of worlds in which A or B is true (the union of their worlds), we can add together the number of worlds in which A is true, plus those worlds where B is true. However, some worlds (those in which both A and B are true) will be counted twice, so we then need to subtract the total of these worlds (the probability $\Pr[A \cap B]$).

Example 2-1 : Random Events

Suppose that we roll a standard, 6-sided die. The event space S encapsulates all possible outcomes, in this case simply which side of the die is up. There are thus 6 possible worlds, each with equal probability 1/6. Two possible events are,

$$A = \text{roll is odd}$$
$$B = \text{roll is 4 or greater.}$$

A *random variable* is a partitioning of the event space into disjoint and exhaustive subsets. For example, suppose we partition S into d subsets, $S = S_1 \cup S_2 \cup \ldots \cup S_d$. Each of these subsets define an event, "S_i", with some probability $\Pr[S_i]$. We can define a random variable $X \in \{1, \ldots, d\}$, such that "$X = i$" corresponds to the event S_i. Because these subsets are disjoint and exhaustive, we have that

$$\sum_{i=1}^{d} \Pr[X = i] = \sum_{i=1}^{d} \Pr[S_i] = \Pr[S] = 1.$$

We call the possible values of the random variable X *states*, the set of all possible values the *domain* of X, and if the number of states d is finite, we call this the *cardinality* of X.

Notationally, we will use $p(X)$ to represent the distribution of X (a probability density or mass function), and, if X is discrete, $p(X = x)$ the probability of the event $X = x$. We will often abuse notation slightly to abbreviate $p(X = x)$ to simply $p(x)$ when the random variable X is clear from the name of the value, x.

Example 2-2 : Bernoulli distribution

A Bernoulli random variable X is a binary-valued random variable, usually $X \in \{0, 1\}$. Let us denote $\Pr[X = 1] = \rho$, where $\rho \in [0, 1]$; since there are only two possible outcomes, we correspondingly have that $\Pr[X = 0] = 1 - \rho$.

We can write the probability distribution of X more mathematically as

$$p(X) = \text{Ber}(X\,;\,\rho) = (\rho)^X\,(1-\rho)^{(1-X)} = \begin{cases} \rho & \text{if } X = 1 \\ (1-\rho) & \text{if } X = 0 \end{cases} \tag{2.1}$$

where the formula gives us an explicit realization of the two possibilities, evaluating to ρ when we plug in $X = 1$ (since $(\rho)^1 = \rho$, and $(1-\rho)^0 = 1$), and evaluating to $1 - \rho$ when we plug in $X = 0$.

There is often more than one way to express $p(X)$ mathematically; for example, the form (2.1) is called the **exponential family** form of the distribution (which we discuss in more detail in Section 2.2). However, since $X \in \{0, 1\}$ we could equivalently write $p(X)$ as,

$$p(X) = (\rho)\,X + (1-\rho)\,(1-X) = \begin{cases} \rho + 0 & \text{if } X = 1 \\ 0 + (1-\rho) & \text{if } X = 0 \end{cases}$$

a pattern sometimes called the **network polynomial** form.

Example 2-3 : Discrete distribution

The most common distribution over a discrete variable is a simple table of probabilities. Let $X \in \{1, \ldots, d\}$ be a d-ary discrete random variable. We can represent the probability distribution of X using a vector ρ, so that $\Pr[X = i] = \rho_i$, with the constraints that all $\rho_i \geq 0$ and $\sum_i \rho_i = 1$. The set of vectors ρ that satisfy these constraints is called the d-dimensional **probability simplex**.

Typically, we will represent ρ using a vector or table of d values. Note that technically, ρ has only $d - 1$ free parameters (since $\rho_d = 1 - \sum_{i<d} \rho_i$); however, it is often convenient to represent ρ using all d values and an equality constraint. As with the Bernoulli distribition, we can explicitly write a formula that evaluates to these probabilites, e.g.,

$$p(X) = (\rho_1)^{\mathbb{1}[X=1]} \cdot (\rho_2)^{\mathbb{1}[X=2]} \cdot \ldots \cdot (\rho_d)^{\mathbb{1}[X=d]}$$

where $\mathbb{1}[\text{expr}]$ is an indicator function that takes value one when the expression 'expr' is true, and zero otherwise.

Joint probabilities

We are rarely interested in only one random variable; when we have more than one variable defined, we typically want to reason about them together. Just as $X = x$ represents the event in which X takes on some particular value x, we can reason about the events corresponding to joint configurations of more than one variable, for example $(X, Y) = (x, y)$, standing in for the event, "$X = x$ and $Y = y$". These joint assignments also form a disjoint and exhaustive partitioning of the event space, and thus their probabilities also sum to one.

Example 2-4 : Dentist Example (R&N)

Suppose that we are going to the dentist, and use random variables to represent the possible outcomes of our visit. We define three variables: let $C = 1$ indicate our having a cavity, $T = 1$ indicate having at toothache, and $D = 1$ indicate the dentist's probe catching on our tooth.

The **joint distribution** $p(T = t, D = d, C = c)$ assigns each possible outcome of the variables, (t, d, c), a probability; in other words, it is a probability distribution over the random variable defined by the tuple (T, D, C). So, each outcome's probability is between zero and one, and the total probability (sum of all entries) is one.

TDC	$p(T, D, C)$
000	0.576
001	0.008
010	0.144
011	0.072
100	0.064
101	0.012
110	0.016
111	0.108

Marginal probabilities

It is often the case that, given a joint distribution over a number of variables, we are interested in the probability distribution of one variable. For instance, we may wish to calculate the probability $\Pr[T = t]$ from the joint distribution $\Pr[T = t, D = d, C = c]$.

The law of total probability tells us that the probability of an outcome, $T = 0$, is simply the probability of all the possible events that are consistent with that outcome. Since the possible outcomes $(T = t, D = d, C = c)$ are exhaustive, the set of worlds with $T = 0$ is simply the union of all worlds consistent with $T = 0$, and

$$\begin{aligned} p(T = 0) &= p\big(\cup_{d,c} T = 0, D = d, C = c\big) \\ &= \sum_{d,c} p(T = 0, D = d, C = c) \\ &= 0.576 + 0.008 + 0.144 + 0.072 \ = \ 0.80 \end{aligned}$$

since, when applying Axiom 4, the outcomes $(T = t, D = d, C = c)$ are also mutually exclusive.

The summation operation $\sum_{d,c}$ is called **marginalizing** the variables D and C. A marginal distribution, such as $p(T)$ or $p(T, D)$, corresponds to the distribution over its arguments (the marginal variables), without reference to (i.e., ignoring the value of) any other variables that might exist.

Conditional probability

Suppose that we observe some information, for example, that we do have a toothache ($T = 1$). We may then wish to reason about the implications of this observation on other events. The conditional probability $p(D|T = 1)$ tells us the probability of the dental probe catching (D) given that we already know we are within one of the worlds in which we have a toothache ($T = 1$).

The rule of conditioning allows us to rewrite a joint distribution in terms of a sequence of conditional distributions, e.g.,

$$p(T, D, C) = p(T)p(D|T)p(C|D, T)$$

This is intuitive: the probability that $T = t$, $D = d$, and $C = c$ all happen can be computed by evaluating the probability that $T = t$ happens, then the probability that $D = d$ also happens given that we already know $T = t$, and so on.

Rearranging, we can view the conditional probability as defined by the ratio,

$$p(D = d|T = t) = p(D = d, T = t)/p(T = t);$$

again, giving the intuitive view that $p(D|T = t)$ is simply a probability distribution on D, but in which we have restricted oursevles to the set of worlds where $T = t$, and the denominator $p(T = t)$ normalizes this distribution to ensure that the total probability over D remains one.

Bayes Rule

A useful consequence of the rule of conditioning is Bayes rule, which tells us how to convert a "forward" or "causal" probability (e.g., the probability that the dentist's probe catches given that we have a cavity) into a "diagnostic" probability, the probability of the underlying cause (a cavity) given our observation (the probe catching):

$$\begin{aligned} p(C = c|D = d) &= \frac{p(D = d|C = c)\, p(C = c)}{p(D = d)} \\ &= \frac{p(D = d|C = c)\, p(C = c)}{\sum_{c'} p(D = d|C = c')p(C = c')} \end{aligned}$$

whenever $p(D = d) \neq 0$. Bayes rule follows trivially from re-arranging two conditional expansions of the joint probability of D, C:

$$p(C = c|D = d)\, p(D = d) \quad = \quad p(D = d, C = c) \quad = \quad p(D = d|C = c)\, p(C = c)$$

then dividing both sides by $p(D = d)$; the second equality then follows by applying the rule of total probability to $p(D = d)$.

Example 2-5 : Bayes rule

Suppose that we wish to compute $p(C = 1|T = 1)$, the probability of having a cavity given that we have a toothache, using Bayes rule. From our general dental knowledge, we have that

$$\begin{aligned} p(T = 1|C = 0) &= 0.1 \qquad & p(C = 0) &= 0.8 \\ p(T = 1|C = 1) &= 0.6 \qquad & p(C = 1) &= 0.2 \end{aligned}$$

Then, if we observe a toothache ($T = 1$), we can compute the diagnostic probability of a cavity as,

$$\begin{aligned} p(C = 1|T = 1) = \frac{p(C = 1, T = 1)}{p(T = 1)} &= \frac{p(T = 1|C = 1)\, p(C = 1)}{p(T = 1|C = 1)p(C = 1) + p(T = 1|C = 0)p(C = 0)} \\ &= \frac{(0.6) \cdot (0.2)}{(0.6) \cdot (0.2) + (0.1) \cdot (0.8)} \quad = \quad 0.6 \end{aligned}$$

Bayes rule is often useful in practice, because it may be easier to specify or estimate some conditional probabilities than others. For example, the probability that the probe catches when there is or is not a cavity is likely something that the dentist can estimate, based on her experience with many patients. On the other hand, $p(C)$ may be different for different patients, depending on their brushing habits, for example. Bayes rule lets us calculate the relatively universal $p(D|C)$, and then apply it to our current belief $p(C)$ for an individual, to obtain a **posterior probability** $p(C|D = d)$ (the probablity our patient has a cavity, after incorporating our observation that $D = d$).

Table-based representations and manipulation. It will often be convenient to represent these functions of discrete random variables as tables, in which case many of the operations that we need to apply (marginalizing, normalizing, etc.) can be captured using basic arithmetic operators over the tables.

Example 2-6 : Table-based computation

Let us re-compute $p(C = 1|T = 1)$, as in Example 2-5, expressing our computation in terms of operations performed on probability tables. Taking the full table of $p(T, D, C)$, we can: (a) *extract* the sub-table in which $T = 1$, corresponding to the function $p(T = 1, D, C)$; (b) sum over the possible values of D, giving $p(T = 1, C)$; (c) normalize this table to sum to one, giving $p(C|T = 1)$.

TDC	$p(T, D, C)$
000	0.576
001	0.008
010	0.144
011	0.072
100	0.064
101	0.012
110	0.016
111	0.108

$\Longrightarrow$ (a)

DC	$p(T = 1, D, C)$
00	0.064
01	0.012
10	0.016
11	0.108

$\Longrightarrow$ (b)

C	$p(T = 1, C)$
0	0.064 + 0.016 = 0.080
1	0.012 + 0.108 = 0.120

$\Longrightarrow$ (c)

C	$p(C\|T = 1)$
0	0.08 / 0.2
1	0.12 / 0.2

Expectations

The expectation of a function $f(\cdot)$ tells us the average, or expected, value of $f(X)$ for a random variable X. For a discrete random variable X, the expectation of f is simply

$$\mathbb{E}_{p(X)}[f(X)] = \sum_x p(X = x) f(x);$$

we often abbreviate this as $\mathbb{E}_X[f]$ or even $\mathbb{E}[f]$ when the random variable X and its distribution $p(X)$ are clear from context.

Many quantities that we would like to predict or reason about can be framed as expectations. For example, if X is a Bernoulli random variable, then

$$\mathbb{E}[X] = \sum_x p(x)x = p(X = 0) \cdot 0 + p(X = 1) \cdot 1 = p(X = 1).$$

In other words, the expected value (sometimes called the *first moment*) of a Bernoulli variable X is simply the probability that X is true.

Expectations are linear functions, so that if a, b are non-random real numbers, we have for any functions f, g:

$$\mathbb{E}[a\, f(X) + b\, g(X)] = a\, \mathbb{E}[f(X)] + b\, \mathbb{E}[g(X)].$$

Linearity is a useful property for analyzing and proving properties about expectations.

Independence

Two random variables X,Y are said to be independent if

$$p(X = x, Y = y) = p(X = x)\, p(Y = y) \qquad \forall x, y$$

or, more compactly, $p(X, Y) = p(X)\,p(Y)$. For two independent variables, observing the value (outcome) of one variable does not influence the probability of the other, since

$$p(X|Y = y) = p(X, Y = y)/p(Y = y) = p(X)p(Y = y)/p(Y = y) = p(X)$$

and similarly for $p(Y|X = x)$.

Independence also simplifies the joint probability distribution: if X and Y are d-ary discrete variables, then (X, Y) has d^2 possible outcomes, and $p(X, Y)$ can be specified by $d^2 - 1$ probabilities. However, if we know that X and Y are independent, these $d^2 - 1$ probabilities are completely determined by only $2d - 2$ values: $d - 1$ for $p(X)$, and $d - 1$ for $p(Y)$.

Example 2-7 : Independence

Let X be a (biased) coin flip, $X \in \{0, 1\}$, and $Y \in \{1, \ldots 4\}$ a (weighted) die roll. If these variables are independent of one another (the outcome of one does not influence the probability of the other), then the joint probability is purely a function of the individual, marginal probabilities, for example:

$p(X)$	$X = 0$	$X = 1$
	0.7	0.3

$p(Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$Y = 4$
	0.2	0.3	0.4	0.1

$\Rightarrow$

$p(X, Y)$	$X, Y = 0, 1$	$X, Y = 0, 2$	$\ldots$	$X, Y = 1, 4$
	$(0.7)(0.2)$	$(0.7)(0.3)$	$\ldots$	$(0.3)(0.1)$

Conditional independence

It is rare that the random variables we are interested in will be completely independent of one another. However, a more common property in practice is that of *conditional independence*, meaning that our variables are dependent only through some mediating quantity. For example, having a toothache and the dental probe catching are not independent events: both are likely caused by a cavity, and so both events will tend to co-occur. But, if we knew definitively whether a cavity was present, it is unlikely that information about a toothache would help predict whether the probe will catch.

Formally, we say two variables X and Y are *conditionally independent* given Z when

$$p(X = x, Y = y|Z = z) = p(X = x|Z = z)\,p(Y = y|Z = z) \qquad \forall x, y, z \tag{2.2}$$

or, more compactly, $p(X, Y|Z) = p(X|Z)\,p(Y|Z)$. As with independence, this means that observing the outcome of, say, Y has no *additional* information about X, beyond what is already provided by Z:

$$p(X|Y = y, Z = z) = p(X|Z = z)p(Y = y|Z = z)/p(Y = y|Z = z) = p(X|Z = z)$$

The definition of conditional independence requires that the relationship (2.2) holds for *all* values of Z. When it is only true for some values z, we say that X and Y are independent in the "context" $Z = z$, called *context specific independence*.

Example 2-8 : Conditional independence, dentist

Conditional independence can be difficult to detect from the joint probability values, but is obvious from evaluating the appropriate conditional probability tables. For example, in the dentist model, it is not obvious from the values of $p(T, D, C)$ that any variables are conditionally independent; however, if we evaluate the conditional distribution,

$$p(D|C,T) = \frac{p(T,D,C)}{p(C,T)} = \frac{p(T,D,C)}{\sum_d p(T, D=d, C)},$$

we find the table values shown at right, from which it is visually apparent that $p(D|C, T = t)$ is identical for all values of t.

T	D	C	$p(D\|C,T)$
0	0	0	0.800
0	0	1	0.100
0	1	0	0.200
0	1	1	0.900
1	0	0	0.800
1	0	1	0.100
1	1	0	0.200
1	1	1	0.900

2.2 Continuous random variables

Sometimes we will be interested in modeling systems of random variables that do not take on values from a discrete set; the most common example are random variables that are real-valued, $X \in \mathbb{R}$. In this setting, we define a probability *density* function $p(x)$ with $p(x) \geq 0$ for all x and $\int p(x)\,dx = 1$. The density function defines the probability of any event $X \in A \subseteq \mathbb{R}$ by $\Pr[X \in A] = \int_A p(x)\,dx$, so that $\Pr[A] \in [0, 1]$.

Example 2-9 : Uniform distribution

For a continuous-valued random variable X defined in some interval $[0, T]$, the *uniform* distribution makes all possible values of X equally likely:

$$p(x) = \begin{cases} 1/T & \text{if } x \in [0,T] \\ 0 & \text{otherwise} \end{cases}$$

Then,

$$\int p(x)\,dx = \int_0^T p(x)\,dx = T \cdot (1/T) = 1.$$

Notice that, unlike discrete distributions, the value of $p(x)$ may be greater than 1, if only in a small region; it is the *integral* of $p(x)$ that cannot be greater than 1. For example, the uniform distribution on interval $[0, \frac{1}{2}]$ has $p(x) = 2$ on that interval, and zero elsewhere.

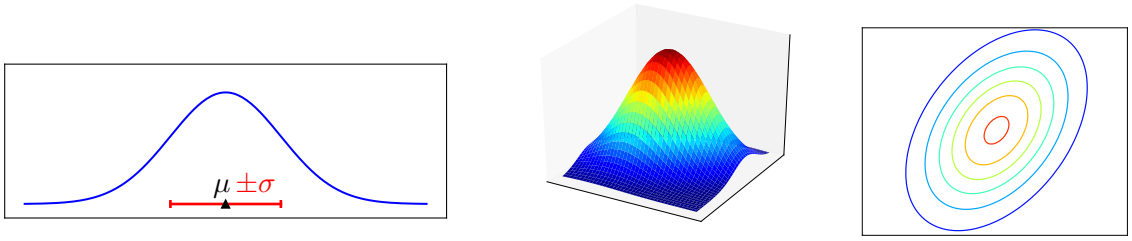
For expectations over continous random variables $X \in \mathbb{R}^n$, we integrate over the outcomes of X, weighted by the density function $p(x)$:

$$\mathbb{E}_{p(X)}[f(X)] = \int p(x) f(x)\,dx.$$

Gaussian distributions

Another important and common continuous distribution is the Gaussian distribution. For a real-valued, scalar (one dimensional) random variable X, the Gaussian distribution is defined by a mean μ and variance σ^2 and is given by

$$p(x) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\Big[-\frac{1}{2}(x-\mu)^2/\sigma^2\Big]$$



(a) (b) (c)

Figure 2.1: (a) Probability density function for a Gaussian distribution in one dimension, with mean μ and standard deviation (spread) σ. (b)–(c) A Gaussian distribution over two variables, $X = [X_1, X_2]$ can be visualized as either a surface, or as a contour plot (where red indicates higher probability and blue lower probability).

The Gaussian distribution is sometimes called the **normal distribution**, and has a classic bell-curve shape centered at μ and with a spread determined by σ (which is required to be positive).

The Gaussian distribution has a natural extension to a distribution over vectors $X \in \mathbb{R}^n$, as well:

$$p(x) = \mathcal{N}(x; \mu, \Sigma) = (2\pi)^{-\frac{n}{2}} |\Sigma|^{-\frac{1}{2}} \exp\Big[-\frac{1}{2}(x-\mu)\Sigma^{-1}(x-\mu)^T \Big]$$

As in the univariate case, the distribution is bell-shaped and centered at the mean vector $\mu \in \mathbb{R}^n$, with a shape and spread that are determined by the $n \times n$ covariance matrix Σ. We require that Σ be symmetric and positive definite, i.e., $v\Sigma v^T > 0$ for any v. This ensures that Σ is invertible, and that its determinant $|\Sigma|$ is positive.

Beta and Dirichlet distributions

Another useful continuous distribution is the Beta distribution, defined for real-valued variables on the domain $[0, 1]$ as,

$$p(x) = \text{Beta}(x; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\, x^{a-1}\,(1-x)^{b-1}$$

where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$ is the standard gamma function; the ratio of gamma functions serves to ensure that $p(x)$ integrates to one.

When $a = b = 1$, it is easy to see that $\text{Beta}(x; 1, 1) \equiv 1$ for all $x \in [0, 1]$, so that $\text{Beta}(x)$ is a uniform distribution. When $a, b > 1$, the beta distribution is unimodal, with its peak at $\frac{a-1}{a+b-2}$, while if a or $b < 1$ the density becomes unbounded at zero or one, respectively. Some example Beta distribution shapes are shown in Figure 2.2.

The Dirichlet distribution generalizes the Beta distribution to vectors x with unit sum, i.e., $x = [x_1, \ldots, x_n]$ such that $\sum_j x_j = 1$. Then, the domain of x is the unit simplex, an $n-1$ dimensional convex polytope whose vertices are the unit directions, $\{e_1 = [1, 0, \ldots 0], \ldots, e_n = [0, \ldots, 0, 1]\}$. The Dirichlet distribution is,

$$p(x) = \text{Dir}(x; \alpha) = \frac{\Gamma\big(\sum_{j=1}^n \alpha_j\big)}{\prod_{j=1}^n \Gamma(\alpha_j)} \prod_{j=1}^n x_j^{\alpha_j - 1}$$

Clearly, when $n = 2$, we have $\text{Dir}(x; \alpha) = \text{Beta}(x_1; \alpha_1, \alpha_2)$ since $x_2 = 1 - x_1$.

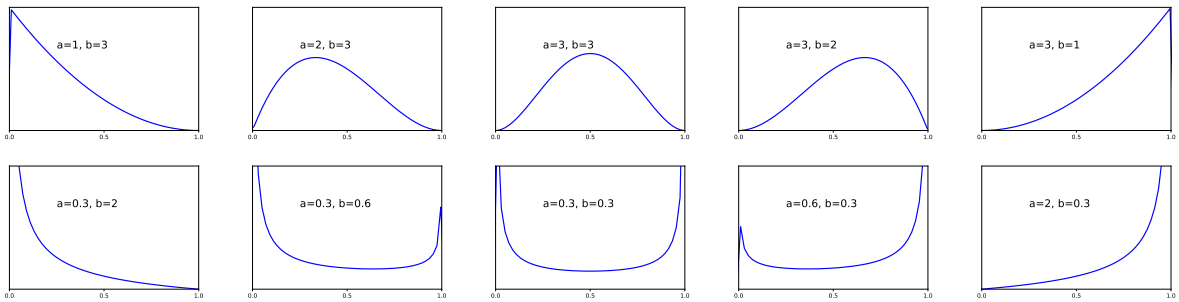
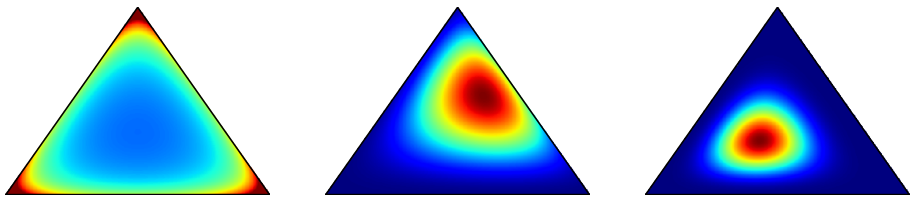


Figure 2.2: Beta distributions (defined for $x \in [0, 1]$) for various values of parameters a, b. When either a or b is less than one, the probability density can be infinite at the boundary (zero or one), although its integral over any interval will always be less than one.



$\alpha = [.3, .3, .3]$ $\alpha = [2, 3, 4]$ $\alpha = [8, 6, 6]$

Figure 2.3: Dirichlet distributions over the unit simplex of dimension $n = 3$ (i.e., $x = [x_1, x_2, x_3]$ with $\sum_j x_j = 1$), for various values of the parameters $\alpha = [\alpha_1, \alpha_2, \alpha_3]$. Like the Beta distribution, when α_i is less than one the density can approach infinity at the boundary of the domain.

We can visualize the shape of the Dirichlet distribution with $n = 3$ for various α, by plotting the unit simplex as the interior of a triangle whose vertices represent the values $[1, 0, 0]$, $[0, 1, 0]$, and $[0, 0, 1]$. Then, like the Beta distribution, when $\alpha_j > 1$ for all j the distribution is unimodal with its peak at $x = (\alpha - 1)/(\sum_j \alpha_j - n)$; when any of the $\alpha_j < 1$ the distribution becomes unbounded somewhere along the boundary. Some examples are shown in Figure 2.3, where blue colors indicate lower probabilities and red indicate higher probabilities.

The exponential family

The distributions we have discussed (e.g., Discrete, Gaussian, Dirichlet, etc.) are all examples of a class of distributions called the *exponential family*, which include any distribution that can be written as

$$p(x; \theta) = \frac{1}{Z(\theta)} \exp\big(\eta(\theta) \odot \phi(x)\big)\, h(x)$$

where $\odot$ is the dot product between the vectors $\eta(\theta)$ and $\phi(x)$, and the value $Z(\theta)$ normalizes the distribution over x. The function $h(x)$ can be used to absorb a change of measure, so that if X's distribution is exponential family, then any invertible transform $f(X)$ also has an exponential family distribution (see, e.g., Section 2.6).

The elements of the vector $\phi(x)$ are called the *features* of the exponential family distribution, while $\eta(\theta)$ transforms the parameters θ into a vector of the same length as $\phi(x)$. If $\eta(\theta) = \theta$, the exponential family is said to be in *canonical* or *natural* form.

Example 2-10 : Bernoulli exponential family form

The Bernoulli distribution is in the exponential family, since for $X \in \{0, 1\}$,

$$\rho^X \, (1-\rho)^{(1-X)} = \exp\Big(\log(\rho)X + \log(1-\rho)(1-X)\Big) = (1-\rho) \, \exp\left(\log\left(\frac{\rho}{1-\rho}\right)X\right)$$

with feature $\phi(X) = X$. We can write an identical distribution using the natural parameter η,

$$p(X;\eta) = \frac{1}{1+\exp(\eta)} \, \exp\big(\eta \, X\big)$$

with the two parameterizations being related through the log-odds and logistic transforms:

$$\eta = \log\left(\frac{\rho}{1-\rho}\right) \qquad \Leftrightarrow \qquad \rho = \frac{1}{1+\exp(-\eta)}$$

The exponential family form highlights the fact that "equivalent" probability distributions can be parameterized in many different ways. As in the Bernoulli example, two common types of parameterizations are so-called "moment" forms, in which we specify a collection of probabilities or expectations that characterize the distribution (parameter ρ), versus forms in which we specify parameters that have useful mathematical properties, such as having the log-linear form of the canonical exponential family (parameter η). Parameterization choices can also lead to different constraints on the valid settings of the model parameters; for example, we require that $\rho \in [0, 1]$, but any value of $\eta \in \mathbb{R}$ produces a valid probability distribution.

Some parameterizations of a distribution may be *over-parameterized*, meaning that multiple settings of the parameters may induce the same probability distribution. For example, we can define a version of the Bernoulli distribution that uses two parameters:

Example 2-11 : Bernoulli two-parameter form

Let random variable $X \in \{0, 1\}$ be defined by the distribution,

$$p(X;\eta_0,\eta_1) = \frac{1}{\exp(\eta_0)+\exp(\eta_1)} \, \exp\big(\eta_1 \, X + \eta_0(1-X)\big).$$

Now, we can see that

$$p(X=1;\eta_0,\eta_1) = \rho = \frac{\exp(\eta_1)}{\exp(\eta_0)+\exp(\eta_1)} = \frac{1}{1+\exp(\eta_0-\eta_1)}.$$

Since this probability completely characterizes the Bernoulli distribution, any settings of (η_0, η_1) that have the same value for $\eta_1 - \eta_0$ will produce an identical probability distribution.

2.3 Learning and parameter estimation

In practice, we often do not know the probabilities that govern a system, but instead must *learn* how that system behaves. To learn, we observe data arising from the system (for example, counting the number of patients at the dentist that have toothaches or probe catches and cavities), and then estimate the probability distribution of the data. In practice, this means that we will select a parameterized class of models for the system (for example probability tables for discrete data, or a Gaussian distribution for continuous-valued data), and then estimate the parameters of our model using our observations.

Frequentist versus Bayesian perspectives

The two major philosophical frameworks for probability are the *frequentist* and *Bayesian* viewpoints. These two frameworks associate different meanings with the notion of probability, resulting in different perspectives that follow the same axioms described in Section 2.1, but which have far-ranging implications on how building a model (e.g., learning from data) should be approached.

From the frequentist perspective, probability is *defined* to be the long-term frequency in which an event happens. So for example, with a coin flip, the probability of heads, $p(H)$, is equal to the rate at which we would observe heads if we repeated a coin flip experiment infinitely many times. If we are handed such a coin, it has an inherent "true" probability, and our task is to estimate that frequency from a more limited observation of flips.

The Bayesian perspective, on the other hand, defines probability to mean a *degree of belief* in an event. This has some significant advantages, in that it allows us to talk about the probability of events that can only happen once ("What is my probability of passing the exam next week?"), or are deterministically true or false but we are ignorant of their value ("What's the probability I left the oven on?"). However, this also means that the probabilites we compute are all relative to our own assumptions and beliefs, which must then be carefully assessed and clearly stated for our results to have meaning to others.

Both perspectives are useful in different settings, and in the sequel we describe what these viewpoints mean for estimating a model from data.

The likelihood function

Likelihood measures the probability of an observation or set of observations under a probability model.

Suppose that we observe the outcome of a random variable X over m independent trials; we will denote our observed data, consisting of m observations, by $D = \{x^{(1)}, \ldots, x^{(m)}\}$. Any model $p(X)$ may be more or less compatible with these observations. The principle of *maximum likelihood* states that we should prefer the model under which our observations look most probable.

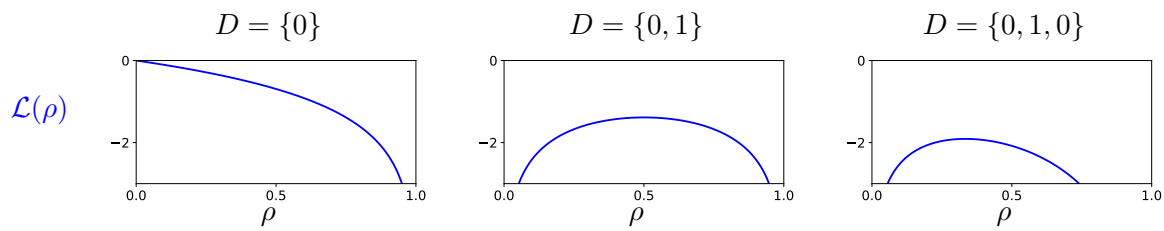
More formally, suppose we have a class of probability distributions $p(X\,;\,\theta)$ parameterized by a vector θ. The log-likelihood of our m observations is

$$\mathcal{L} = \log p(D\,;\,\theta) = \log \prod_i p(X = x^{(i)}\,;\,\theta) = \sum_i \log p(X = x^{(i)}\,;\,\theta)$$

where the second equality holds due to the independence of the $x^{(i)}$ in our data D. Since the log-likelihood $\mathcal{L}$ depends on the value of the parameters θ, we can denote this dependence explicitly as $\mathcal{L}(\theta)$; from this perspective, we can vary θ and see how well different values explain our observations D. We illustrate the likelihood function with a few examples.

Example 2-12 : Bernoulli likelihood

Suppose that we observe a set of coin flips, with $X = 1$ indicating heads, and $X = 0$ tails. Then, we can evaluate the probability of our data set D as a function of the model's parameter ρ, the probability of observing heads. If we observe only one coin flip, $D = \{0\}$, then, when $\rho = 0$ (a trick coin, that always comes up tails), we have $\mathcal{L} = \log(1) = 0$. As $\rho \to 1$ (our coin is supposedly more and more likely to come up heads), our observed outcome in D looks increasingly improbable.



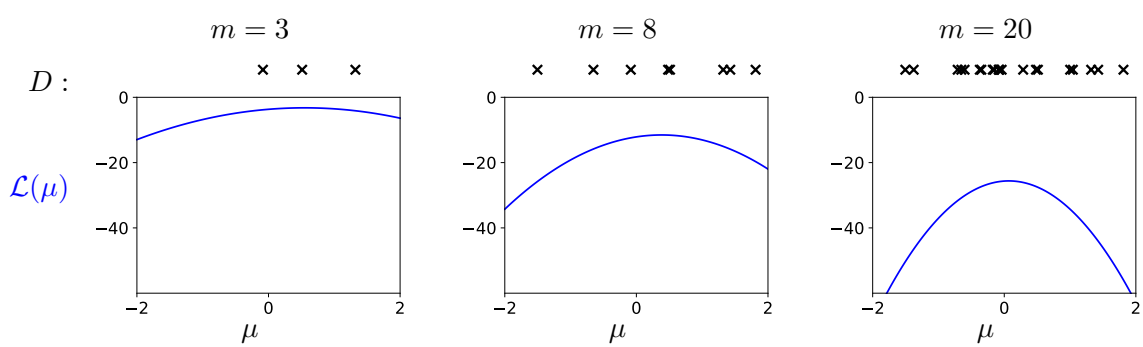
If we observe more data, the likelihood will change. for example, if we observe two coin flips $D = \{0, 1\}$ (one tails, one heads), now the coin cannot have $\rho = 0$, since then we would never have observed a heads outcome; the log-likelihood of our two data points is $\mathcal{L} = \log \rho + \log(1 - \rho)$, which is highest when $\rho = 1 - \rho = \frac{1}{2}$.

Since for discrete variables, the probability of any observation is less than one, the log-likelihood tends to become smaller quickly as we observe more data. Thus it is sometimes common to examine or report the average log-likelihood value, $\frac{1}{m}\mathcal{L}$.

We can similarly evaluate the log-likelihood of a dataset D of continuous variables, in which case $p(X\,;\,\theta)$ is a density rather than a probability mass, but otherwise the principle is the same. Given the values of $x^{(i)} \in D$, different settings of the parameters θ are more or less compatible with the observations:

Example 2-13 : Gaussian likelihood

Consider the likelihood function of a Gaussian model with mean parameter μ (and fixed variance $\sigma^2 = 1$). Since our observations are now real-valued, we plot them along the top of the figure, using the same scale as the value of μ.



Again, when we have only a few data points, many different values of μ are relatively plausible, since (for example) it is easy to be unlucky and draw three samples in a row that are all larger than μ. However, as m increases, it becomes less and less plausible that μ is not near the center of the observed samples.

Maximum likelihood estimation

A simple way to construct a probability distribution that explains our observed data D is to use the principle of maximum likelihood, i.e., to select the model that maximizes the log-likelihood of our observations. Generally speaking, this will be the model that is most consistent with the observed data, in the sense that it makes the data appear as likely as possible.

Finding maximum likelihood estimates

In many models, it is possible to derive the maximum likelihood estimate (MLE) of the parameters directly, in closed form. For example, consider a Bernoulli distribution:

Example 2-14 : Maximum likelihood estimation: Bernoulli

The log-likelihood of the Bernoulli distribution for a collection of m i.i.d. samples $D = \{x^{(1)}, \ldots, x^{(m)}\}$ is given by,

$$\mathcal{L} = \sum_{i=1}^{m} \log(\rho)\mathbb{1}[x^{(i)} = 1] + \log(1-\rho)\mathbb{1}[x^{(i)} = 0] \qquad = m_1 \log(\rho) + m_0 \log(1-\rho)$$

where m_0, m_1 are the number of data in D with $x^{(i)}$ equal to zero and one, respectively. Taking the derivative with respect to ρ and setting it equal to zero[1], we find that $\mathcal{L}$ has a unique maximum at,

$$\frac{\partial \mathcal{L}}{\partial \rho} = m_1 \frac{1}{\rho} - m_0 \frac{1}{1-\rho} = 0 \qquad \Rightarrow \qquad \rho = \frac{m_1}{m_0 + m_1} = \frac{m_1}{m},$$

the empirical fraction of ones in the data set. Indeed, this matches the maxima we observed for our Bernoulli data sets in Example 2-12.

When our probability distribution has more than one parameter, the maximum likelihood estimator is given by optimizing over all parameters jointly. A simple example is the one-dimensional Gaussian distribution:

Example 2-15 : Maximum likelihood estimation: Gaussian

For a Gaussian distribution with mean μ and variance $\nu = \sigma^2$, the log-likelihood is,

$$\begin{aligned}\mathcal{L} &= -\frac{1}{2}\sum_{i=1}^{m}\Big[\log(2\pi\nu) + (x^{(i)} - \mu)^2/\nu\Big] \\ &= -\frac{m}{2}\log(2\pi\nu) - \frac{1}{2\nu}\sum_i \left(x^{(i)}\right)^2 + \frac{2\mu}{2\nu}\sum_i x^{(i)} - \frac{m}{2\nu}\mu^2\end{aligned}$$

[1]Technically, we should take care to check that this is indeed a maximum (and not a minimum or plateau), and examine the boundary of the domain of ρ (which may contain the optimum without it having derivative zero), but we omit these details here.

Taking the derivatives with respect to μ and ν gives,

$$\frac{\partial \mathcal{L}}{\partial \mu} = \frac{1}{\nu}\sum_i x^{(i)} - \frac{m}{\nu}\mu \qquad\qquad \frac{\partial \mathcal{L}}{\partial \nu} = -\frac{m}{2}\frac{1}{\nu} + \frac{1}{2\nu^2}\sum_i \left(x^{(i)} - \mu\right)^2$$

Setting both equations to zero and solving them simultaneously, we find

$$\hat{\mu} = \frac{1}{m}\sum_i x^{(i)} \qquad \text{and} \qquad \hat{\nu} = \frac{1}{m}\sum_i \left(x^{(i)} - \hat{\mu}\right)^2$$

i.e., the MLE is given by the empirical mean and variance of the data[2].

Some parameterizations of distributions may involve constraints; for example, the discrete distribution defined by $\rho = [\rho_1, \ldots, \rho_d]$ in Example 2-3 requires that $\sum_i \rho_i = 1$. If necessary, we can incorporate such constraints using Lagrangian optimization, in which we optimize an objective that includes both the log-likelihood $\mathcal{L}$ and terms corresponding to the constraints, multiplied by Lagrange multipliers λ. We illustrate with an example:

Example 2-16 : Maximum likelihood estimation: discrete

Suppose we have a discrete distribution $p(X)$ over d possible outcomes $\{1, \ldots, d\}$, parameterized by $\rho = [\rho_1, \ldots, \rho_d]$ as in Example 2-3. We observe m data $\{x^{(i)}\}$, and group them by their outcome value x, i.e., m_1 observations of $X = 1$, m_2 observations of $X = 2$, and so forth. Then, the log-likelihood is,

$$\mathcal{L} = \sum_i \log \rho_{x^{(i)}} = \sum_x \log \rho_x^{m_x}$$

subject to the constraint $\sum_x \rho_x = 1$. We then optimize the Lagrangian saddle point optimization,

$$\max_\rho \min_\lambda \tilde{\mathcal{L}}(\rho, \lambda) \qquad \text{where} \quad \tilde{\mathcal{L}}(\rho, \lambda) = \sum_x \log \rho_x^{m_x} \; + \; \lambda\big(1 - \sum_x \rho_x\big)$$

Intuitively, maximizing over ρ with λ fixed will attempt to increase all the ρ_x; but then, $\sum_x \rho_x > 1$, and the minimization over λ will increase λ, penalizing for the larger ρ_x. At the saddle point, these two forces will balance, enforcing the constraint. The saddle point is defined by the simultaneous solution to

$$\forall x, \quad \frac{\partial \tilde{\mathcal{L}}}{\partial \rho_x} = \frac{m_x}{\rho_x} - \lambda = 0 \qquad \text{and} \qquad \frac{\partial \tilde{\mathcal{L}}}{\partial \lambda} = (1 - \sum_x \rho_x) \qquad = 0$$

which is simply $\rho_x = m_x/m$, the empirical probabilities of the outcomes x.

The maximum likelihood parameter estimate has a number of appealing properties. First, from the perspective of frequentist statistics, maximum likelihood estimation is *asymptotically consistent*, which means that if our data set is drawn from a distribution with the same parametric form $p(X\,;\,\theta)$ and true parameter value θ^*, our maximum likelihood estimate $\hat{\theta}_{\text{MLE}}(D)$ will approach the true θ^* as our dataset

[2] In some cases, users may prefer to use the also-common estimator $\hat{\nu} = \frac{1}{m-1}\sum_i (x^{(i)} - \hat{\mu})^2$, which is an *unbiased* estimator of the variance; as m becomes large, the two estimators become indistinguishable.

D increases in size ($m \to \infty$). It is also asymptotically normal, meaning that the distribution of $\hat{\theta}_{\text{MLE}}(D)$ (over many draws of data sets D) approaches a Gaussian distribution as m increases, and finally that it is optimally efficient (has the smallest variance) among all asymptotically normal estimators (Fisher, 1925).

Another nice property of the maximum likelihood estimator is that it is invariant to transformations in the parameterization. Intuitively, if the maximum of a function $\mathcal{L}(\eta)$ is at η^*, and we have any function $\eta(\theta)$, then any point θ^* such that $\eta(\theta^*) = \eta^*$ is a maximum of $\mathcal{L}(\eta(\theta))$. To see why this is useful, consider the Bernoulli distribution. We saw that the MLE estimate of the moment parameter ρ was the empirical mean, $\hat{\rho}_{\text{MLE}} = \frac{m_1}{m}$. But, suppose that instead, we want to use the cannonical form, in terms of the natural parameter η? We can use the invariance property to easily derive that the MLE estimate of η is simply the transformation of $\hat{\rho}_{\text{MLE}}$, i.e.,

$$\hat{\eta}_{\text{MLE}} = \log\left(\frac{\hat{\rho}_{\text{MLE}}}{1 - \hat{\rho}_{\text{MLE}}}\right) = \log(m_1/m_0).$$

In fact, we have already used this property when deriving the MLE of the Gaussian distribution in Example 2-15: we found the optimal value of the variance, $\nu = \sigma^2$, from which we can compute the standard deviation, $\hat{\sigma}_{\text{MLE}} = \sqrt{\hat{\nu}_{MLE}}$.

Maximum likelihood and the exponential family

The intuitive "empirical" MLE estimators of the Bernoulli and Gaussian models are examples of a more general property of exponential family models, that the MLE condition corresponds to matching empirical expectations from the data set. Specifically, suppose we have a canonical exponential family model $p(X\,;\,\theta)$. Then, the log-likelihood of m i.i.d. observations is,

$$\mathcal{L} = -m \log Z(\theta) + \sum_i \sum_j \theta_j \phi_j(x^{(i)}) + \sum_i \log h(x^{(i)})$$

Now, consider the derivative of $\log Z(\theta)$ with respect to θ. We have

$$Z(\theta) = \int \exp\Big(\sum_j \theta_j \phi_j(x)\Big)\, h(x) dx$$

$$\Rightarrow \qquad \frac{\partial \log Z(\theta)}{\partial \theta_k} = \frac{1}{Z(\theta)} \int \exp\Big(\sum_j \theta_j \phi_j(x)\Big)\, \phi_k(x)\, h(x) \quad = \mathbb{E}[\phi_k(x)] \tag{2.3}$$

In other words, the derivative of the log partition function, $\log Z(\cdot)$, with respect to parameter θ_k is simply the expected value of feature ϕ_k under the current model $p(X\,;\,\theta)$. Using this fact to solve for the stationary points of the log likelihood, we have

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = 0 \qquad \Leftrightarrow \qquad \mathbb{E}[\phi_k] = \frac{1}{m} \sum_i \phi_k(x^{(i)}),$$

i.e., at the maximum likelihood setting of θ, we should find that the model's estimates of the expectations $\mathbb{E}[\phi_k]$ exactly match the empirical averages $\frac{1}{m}\sum \phi_k(x^{(i)})$ for each of the exponential family's features ϕ_k. This is sometimes called a *moment-matching* condition.

We can also see that the average log-likelihood, $\frac{1}{m}\mathcal{L}$, depends on the data only through the average statistics of the features ϕ_k, i.e., $\frac{1}{m}\sum_i \phi_k(x^{(i)})$. For this reason, these statistics are the **sufficient statistics** (Fisher, 1922) of the distribution; the value of the sufficient statistics completely characterize the data for the purposes of estimating the model parameters.

Returning to Example 2-14, we see that $\phi(X) = [X]$ and indeed, the MLE parameter estimate ensures that the model has expectation $\mathbb{E}[X] = \hat{\rho} = \frac{1}{m}\sum x^{(i)}$, the empirical probability. Similarly, in Example 2-15, we have $\phi(X) = [X, X^2]$ and can verify that the MLE parameter settings ensure that

$$\mathbb{E}[X] = \hat{\mu} = \frac{1}{m}\sum x^{(i)} \qquad\qquad \mathbb{E}[X^2] = \hat{\nu} + \hat{\mu}^2 = \frac{1}{m}\sum \left(x^{(i)}\right)^2$$

Thus, we can fit a model to our data by simply searching for parameter values that reproduce the empirical moments of our features ϕ_k in our data. This technique is sometimes called the *method of moments*, and can be applied to distributions outside the exponential family as well, although in that case it will not, in general, be equivalent to maximum likelihood estimation.

Maximum likelihood and overfitting

However, from a machine learning perspective, maximum likelihood estimators have a significant drawback: *overfitting*. The maximum likelihood principle tells us that we should select the model that assigns highest probability to our data observations. But as we allow our models to become more complex, we are increasingly able to "explain" even random variations in the (finite) data set. For example, in our Bernoulli model of Example 2-12, after observing only a single coin flip $D = \{0\}$, the maximum likelihood estimate was $\hat{\rho} = 0$, i.e., that the coin was deterministically tails. Although this estimate gives a high probability to the data point we have seen so far, it is probably an unreasonable conclusion given that we have only seen one example; and indeed, it is not consistent with (gives zero probability to) the next data point, $X = 1$.

To see this same effect in a more practical setting, consider the following example of a histogram-like density estimate:

Example 2-17 : Histograms

Suppose we have m real-valued data points, $D = \{x^{(i)}\}$ in some finite domain, say $x^{(i)} \in [0, 1]$. We decide to fit a *histogram*-like distribution consisting of d equally spaced bins, each with a constant probability within the bin:

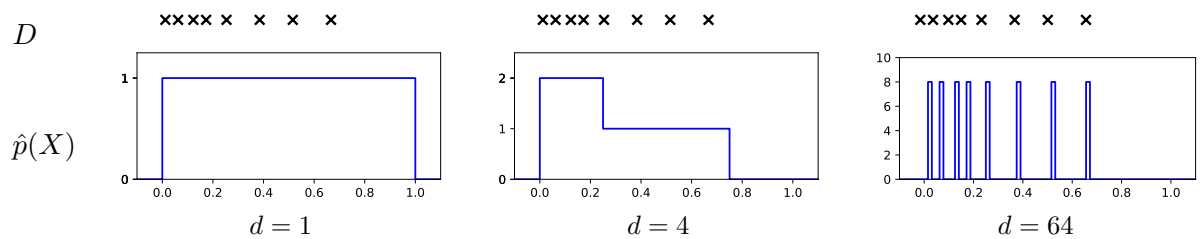
$$p(X\,;\, h = [h_1, \ldots h_d]) = d \sum_{j=1}^{d} h_j \, \mathbb{1}[\tfrac{j-1}{d} \leq X < \tfrac{j}{d}]$$

i.e., we have d parameters in the vector h, with $\sum_j h_j = 1$, and assign each probability mass h_j uniformly across the jth subinterval. It is not difficult to prove that the MLE of h is simply $h_j = \frac{1}{m}\sum_i \mathbb{1}[\frac{j-1}{d} \leq x^{(i)} < \frac{j}{d}]$, the fraction of data which fall into interval j (so that the resulting MLE distribution is simply a normalized d-bin histogram).

When $d = 1$, we have $h_1 = 1$ (i.e., there are no parameters to fit) and $p(X\,;\,[h_1])$ is the uniform distribution, with log-likelihood $\mathcal{L}([h_1]) = 0$. When $d = 2$, if we select $h_1 = h_2 = \frac{1}{2}$, we again obtain the uniform distribution; but if more data are located in $[0, \frac{1}{2}]$ than in $[\frac{1}{2}, 1]$, the maximum likelihood estimator will choose $h_1 > h_2$, giving a higher likelihood than the single-bin model. We say that these models are **nested**: any instance of the simpler model can be expressed using some setting of the parameters of the more complex model. This means that, for nested models, the maximum of the more complex model is always larger than the maximum of the less complex model:

$$\max_{h_1,h_2} \mathcal{L}([h_1, h_2]) \geq \max_{h_1} \mathcal{L}([h_1])$$

If we continue to double the value of d, we are able to potentially increase the log-likelihood, until each data point is in its own individual bin, with $h_j = \frac{1}{m}$ if any data are in bin j and zero otherwise; at this point, the likelihood of the data is $\max_h \mathcal{L}([h_1, \ldots, h_d]) = m \log(d/m)$, which is unbounded as d increases. Intuitively, the bins are becoming smaller and smaller, until they explain only the observed data but with extremely high probability (since each bin is extremely narrow). For example, suppose we have $m = 8$ data points, and fit histograms with increasing d:



For large d, our model explains the data in our observation set D extremely well, but has ceased to be a reasonable predictor of the probability at any points not in D.

In fact, as our number of bins $d \to \infty$, the histogram density approaches the so-called **empirical density** of D,

$$\hat{p}(X\,;\,D) = \frac{1}{m}\sum_i \delta(X - x^{(i)}) \tag{2.4}$$

where $\delta(x)$ is the Dirac delta function, defined so that

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases} \qquad \text{and} \qquad \int_\Omega \delta(x)\,dx = \begin{cases} 1 & \text{if } 0 \in \Omega \\ 0 & \text{otherwise} \end{cases}$$

i.e., $\hat{p}(X\,;\,D)$ places an infinitessimally thin point mass at each data point $x^{(i)}$.

In general, as our models become more complex (have more parameters and flexibility to fit arbitrary probability distributions), the maximum likelihood model will fit the finite set of observations in D better and better, but at some point will cease to *generalize*, i.e., provide reasonable predictions of new data not already in D. In other words, our model will *overfit* to D. Observing more data (increasing m) can reduce this overfitting effect, but is not always possible in practice.

Bias and variance of an estimator; model selection; hold out/cross validation.

Posterior distributions

One way around this overfitting effect is to take the Bayesian perspective on modeling, in which the model itself (i.e., its parameters) are uncertain, and we can evaluate our *belief* about the model given observed data D. In this framework, we have a *prior belief* about what types of distributions we expect to see; then observe D and update our prior given the evidence. If we assume that $p(X)$ comes from some parameterized set of distributions[3], $p(X \mid \theta)$, we can represent our prior over which of these distributions are more or less likely as a prior over the parameters, $p(\theta)$.

[3]In the Bayesian perspective, the parameters θ are themselves random variables, and so we typically use the "conditional probability" notation $p(X|\theta)$ instead of the more frequentist "parameter" notation $p(X\,;\,\theta)$.

Then, our *posterior distribution* over θ given D is,

$$p(\theta \mid D) = \frac{1}{p(D)}\, p(D \mid \theta)\, p(\theta) \qquad\qquad p(D) = \int_\theta p(D \mid \theta)\, p(\theta)\, d\theta \tag{2.5}$$

where $p(D)$ is called the *marginal likelihood* of the data D, and normalizes $p(\theta|D)$ to integrate to one.

Conjugate priors

While (2.5) gives us a general relationship for updating our beliefs about θ given data, in practice, actually representing and computing $p(\theta|D)$ can be difficult. However, if the prior $p(\theta)$ and likelihood $p(D|\theta)$ are well matched, it can be easy. We say that a prior and likelihood are *conjugate* if the posterior, $p(\theta|D)$ can be represented in the same parametric form as the prior.

Example 2-18 : Beta-Bernoulli Conjugacy

Suppose that $p(X|\rho)$ is Bernoulli (with $X \in \{0, 1\}$ and $\rho \in [0, 1]$), and that our prior, $p(\rho)$, takes the form of a Beta distribution: $p(\rho) = \text{Beta}(\rho\,;\, a, b)$. Then, we have

$$\begin{aligned} p(\rho|D) &\propto p(\theta\,;\, a, b) \prod_i p(x^{(i)}|\theta) = (\rho)^{a-1}(1-\rho)^{b-1}\, (\rho)^{x^{(i)}}(1-\rho)^{1-x^{(i)}} \\ &= (\rho)^{(a-1)+m_1}(1-\rho)^{(b-1)+m_0} \\ &\propto \text{Beta}(\rho\,;\, a + m_1, b + m_0) \end{aligned}$$

So, our posterior $p(\rho|D)$ is also a Beta distribution, with updated parameters $a + m_1$ and $b + m_0$.

A similar comparison shows that the Dirichlet prior over a vector ρ is conjugate to the Discrete distribution whose outcome probabilities are given by ρ. In fact, all exponential family distributions have conjugate priors. Additionally, conjugacy means that if the normalizing constants of our conjugate distributions $p(\theta)$ and $p(\theta|D)$ can be evaluated in closed form, we are also able to evaluate the marginal likelihood $p(D)$ in closed form.

Posterior estimators

While the posterior distribution $p(\theta|D)$ characterizes our belief given the data D, i.e., the probability of all possible values of θ, it is sometimes useful to extract a single prediction about the value of θ (a *point estimate*). The two most common are the *posterior mean* (PM) estimate and the *maximum a pesteriori* (MAP) estimate:

$$\begin{aligned} \hat{\theta}_{\text{PM}} &= \mathbb{E}_{p(\theta|D)}[\theta] = \int \theta\, p(\theta|D) d\theta \\ \hat{\theta}_{\text{MAP}} &= \arg\max_\theta \log p(\theta|D) = \arg\max_\theta \log p(D|\theta) + \log p(\theta) \end{aligned}$$

Theoretically, the PM estimate can be viewed as trying to minimize the squared error in our estimate $\hat{\theta}$ under our current belief; it is also computationally convenient since it requires only an expectation over the posterior, which allows it to be estimated using approximations such as Monte Carlo methods (see Chapter 8). On the other hand, the MAP estimate can be viewed as maximizing the log-likelihood $\log p(D|\theta)$, but also taking into account the prior belief about θ in the form of a $\log p(\theta)$ term. From the

machine learning perspective, this is closely related to the techinque of *regularization*, in which "extreme" values of the parameters θ are penalized during learning. In general, as the number of observed data m increases, the magnitude of the likelihood term will eventually dominate the prior, and our estimators will approach the maximum likelihood estimates.

Example 2-19 : Bernoulli posterior estimates

The posterior mean and MAP estimates of a Bernoulli likelihood with prior Beta$(\rho; a, b)$ and observations consisting of m_1 and m_0 data of values 1 and 0, respectively, is,

$$\hat{\rho}_{\text{PM}} = \frac{a + m_1}{a + b + m_1 + m_0} \qquad \hat{\rho}_{\text{MAP}} = \frac{a - 1 + m_1}{a + b - 2 + m_1 + m_0}$$

Both estimators look similar to the MLE estimator, $\frac{m_1}{m_1+m_0}$, but shifted toward the prior distribution's mean, $\frac{a}{a+b}$. When the prior is a uniform distribution ($a = b = 1$), the MAP estimator is exactly equal to the MLE estimator on the data. However, the PM estimator, and MAP estimator for $a, b > 1$, are *regularized* by the prior so that $\hat{\rho}$ will not be exactly zero even if we observe no data with $x^{(i)} = 1$, and similarly is never exactly one; the prior distribution pulls it away from its most extreme values.

Since the prior distribution's parameters a, b play a role equivalent to the observed data counts m_1, m_0, but do not derive from actual data measurements, their values are sometimes referred to as **pseudo-counts**. Many conjugate priors can be viewed as being equivalent to including additional "imagined" data. From this perspective, the total $a + b$, for $a, b > 1$, represents the strength of our prior's belief in the value $\rho = \frac{a}{a+b}$, and how many observations m are required to meaningfully shift our estimate.

Sequential belief updating

The Bayesian viewpoint on updating a prior distribution gives a nice perspective on sequential learning, in which we gradually gather more observations over time. In particular, we start with a prior $p(\theta)$, and observe some data D_1. After observing D_1, our new belief about the model is given by the posterior, $p(\theta|D_1)$. Now, suppose that we are then shown some additional data D_2. Before seeing D_2, our belief is $p(\theta|D_1)$ – therefore this is our current "prior." The data likelihood of D_2 is $p(D_2|\theta, D_1)$[4], giving a posterior $p(\theta|D_1, D_2)$, exactly as if we had observed $D_1 \cup D_2$ at the same time. See Figure 2.4 for a visual illustration of this process.

Hyper-priors

Using conjugate priors can be somewhat restrictive, since from the Bayesian perspective we are supposed to use $p(\theta)$ to encode our prior expectations about the model, which may not fit neatly into the conjugate family of distributions. While it is possible to use non-conjugate priors, this adds significant computational complexity, and we defer discussion to Chapter 8. However, a simpler way to add flexibility to our prior is via **hyper-priors**, in which we express our uncertainty over the parameters of our prior distribution $p(\theta)$.

Example 2-20 : Coin toss hyper-prior

[4] If D_1 and D_2 are conditionally independent given θ, i.e., independent observations of the same system (characterized by θ), this reduces to simply $p(D_2|\theta)$.

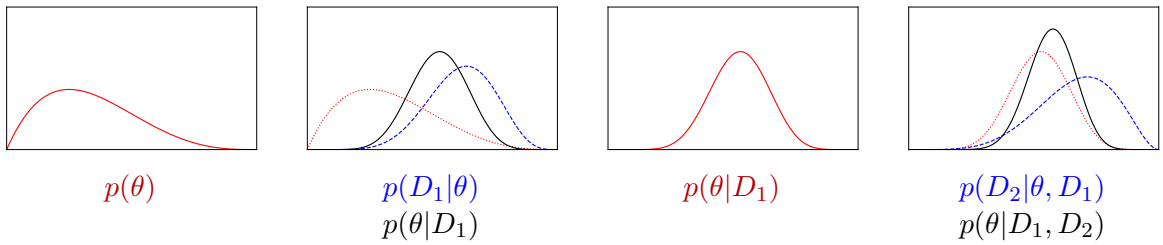
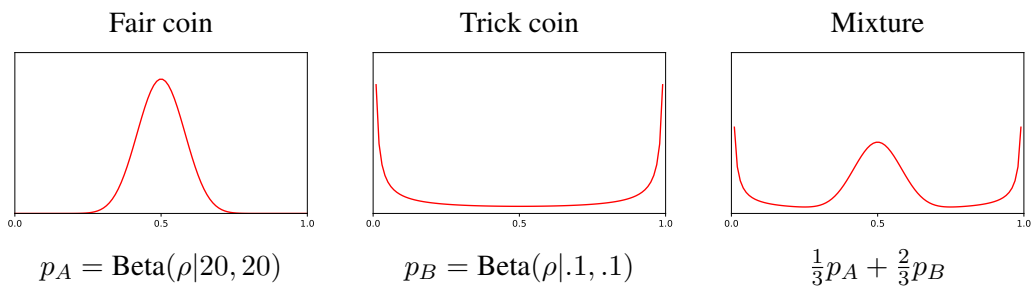


Figure 2.4: Sequential belief updates after observing data D_1, then data D_2. After observing D_1, our posterior $p(\theta|D_1)$ becomes our new "prior" belief for future data, which can then be incorporated via Bayes rule when observed.

Suppose someone asks us to gamble with them on the outcome of a coin flip. We know that arbitrary coins (such as the ones in our own pockets) are pretty close to fair, i.e., have $p(\text{heads}) \approx \frac{1}{2}$. However, we may be suspicious that our opponent is attempting to scam us, and has brought a weighted coin or trick coin with two heads or two tails. Either scenario can be reasonably approximated using a Beta distribution, but our uncertainty as to *which* of these two settings is correct means that our true prior may be a mixture (weighted average) of the two possibilities, e.g.,



The resulting mixture distribution, $\alpha p_A + (1 - \alpha)p_B$, corresponds to placing a hyper-prior over the parameters a, b of the Beta distribution, with uncertainty over which of the two values is correct.

Hyper-priors are not limited to mixture models (we could place a continuous distribution over a and b, for example), nor to only one level (e.g., we could place another prior on the value of α in our mixture prior to express our uncertainty over how likely our opponent is to try to cheat). In practice, we often strike a balance between accurately capturing our true prior belief and remaining computationally convenient.

Weakly informative priors

One complaint about the Bayesian perspective is that our estimates are inherently influenced by our choice of prior distribution. Although the Baysian philosophy argues that this is only natural, in some settings practitioners may attempt to choose a prior that expresses as little information as possible about the parameters. For example, we might choose a uniform prior, to express the idea that all outcomes are equally likely.[5] For a Bernoulli likelihood function, the uniform distribution on the interval $[0, 1]$ is

[5] Other attempts to define weakly informative priors include **maximum entropy priors** (**?**), or **reference priors** (**?**).

given by Beta$(\rho; 1, 1)$, and has the convenient property that its MAP estimate is $\hat{\rho}_{\text{MAP}} = \frac{m_1}{m_1+m_0} = \hat{\rho}_{\text{MLE}}$, the same as the maximum likelihood estimate of ρ. For unbounded parameters, such as the mean μ of a Gaussian distribution, a prior which places equal probability on all possible values, $p(\mu) \propto 1$, cannot be normalized and is thus called an **improper prior**. It is possible to use improper priors in Bayesian inference, at least in settings where the posterior given our observations is guaranteed to be proper. Alternatively, we might choose a weak but proper prior, such as a Gaussian prior with a very large variance, to express our uncertainty about the value of μ.

However, the sometimes-used term **uninformative prior** is something of a misnomer, since all prior distributions express information about the likely parameter values and thus influence our estimates. One way to see this property is by considering equivalent parameterizations of a distribution.

Example 2-21 : Priors for the Bernoulli likelihood

Expressing the Bernoulli distribution in terms of its moment parameter, $\rho = \mathbb{E}[X]$, we can see that the uniform prior, $\rho \sim p(\rho) = \text{Beta}(\rho; 1, 1)$ is, in some sense, uninformative – all values of ρ are equally likely. However, suppose that we decide to represent our Bernoulli distribution using its natural parameter form, $\theta = \log \frac{\rho}{1-\rho}$. Using a transformation of variables (see Section 2.6), we can compute the distribution $p(\theta)$ that is equivalent to the uniform $p(\rho)$; but it is decidely non-uniform, with a peak at $\theta = 0$. Similarly, we can see that the posterior mean estimator is also not invariant to transformation, since

$$\hat{\theta}_{\text{PM}} = \mathbb{E}_p[\theta] = \mathbb{E}_{p(\rho)}\left[\log \frac{\rho}{1-\rho}\right] \neq \log \frac{\hat{\rho}_{\text{PM}}}{1-\hat{\rho}_{\text{PM}}}, \qquad \hat{\rho}_{\text{PM}} = \mathbb{E}_{p(\rho)}[\rho]$$

Thus, our choice of parameterization greatly affects the concept of "uninformative" – are all probability values equally likely, or are all log-odds ratio values equally likely? While the full prior or posterior distribution can typically be transformed to evaluate the distribution of some alternative parameterization, a point estimate derived from these distributions will not be transformation invariant (in contrast to the invariance of the maximum likelihood estimator).

Bayesian model selection

The Bayesian perspective of statistics naturally provides mechanisms to avoid overfitting. In particular, maintaining posterior uncertainty over our model parameters means that when a model is fit with few data points, it naturally remains very uncertain about its predictions. In contrast to using a model's maximum likelihood, we can directly compare models of different complexity in the Bayesian setting using the data marginal log-likelihood,

$$\mathcal{L}_{marg} = \log p(D) = \log \int p(D|\theta)p(\theta)d\theta.$$

Intuitively, as the number of parameters in θ increase, the prior $p(\theta)$ will spread its probability mass over many values of θ. However, only some of the values of θ are consistent with D, i.e., have high likelihood $p(D|\theta)$. If the prior places enough mass on values of θ that explain D well, the marginal likelihood will be high; but if the prior spreads its probability too thinly and includes too many models that are not consistent with D, the marginal likelihood will decrease.

In the case of conjugate priors, the marginal likelihood can be evaluated directly; however, in other settings we must resort to numerical approximations. One simple approximation is the *Bayesian information criterion*, or BIC penalty. The BIC penalty is motivated by taking an asymptotic analysis of a

quadratic approximation to the marginal likelihood around the maximum likelihood estimate; this gives the penalized maximum likelihood value,

$$\mathcal{L}_{\text{BIC}} = \max_{\theta} \log p(D|\theta) - \frac{d}{2} \log m$$

where d is the number of parameters in the model (e.g., the length of the parameter vector θ). Notice that, since D consists of m i.i.d. measurements, the log-likelihood $\log p(D|\theta)$ grows linearly with m, and so will dominate the penalty term for sufficiently large m. The BIC penalty is convenient for a "pseudo-Bayesian" approach, since it requires only the maximum likelihood point estimate of θ, and the number of parameters in the model, but not an explicit prior $p(\theta)$ or computation of the marginal likelihood. Thus the BIC, and related complexity-penalized likelihood scores such as the Akaike information criterion (Akaike, 1973) can be used to do model selection in frequentist statistics models as well.

Example 2-22 : Bayesian histogram estimator

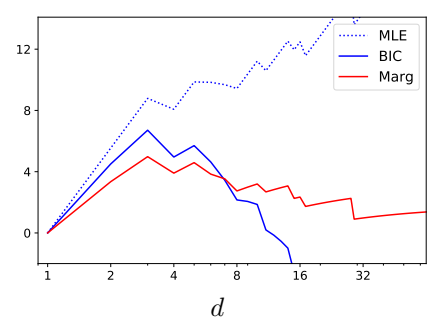
Consider the data and histogram estimator from Example 2-17. Since the parameter vector h sums to one, we can easily apply a Dirichlet prior, $p(h) = \text{Dir}(h; \alpha)$ for h. Now, using the normalization constants for the Dirichlet distribution, it is straightforward to derive the marginal log-likelihood for D under this prior:

$$\mathcal{L}_{marg} = m \log d + \log \frac{\Gamma(\sum_j \alpha_j)}{\prod_j \Gamma(\alpha_j)} - \log \frac{\Gamma(m + \sum_j \alpha_j)}{\prod_j \Gamma(m_j + \alpha_j)}$$

where d is the number of bins, and m_j is the number of data which fall into bin j. Similarly, since the maximum likelihood estimate for h_j is $h_j = \frac{m_j}{m}$, the BIC-penalized log-likelihood is,

$$\mathcal{L}_{\text{BIC}} = \sum_j m_j \log \frac{d\, m_j}{m} - \frac{d}{2} \log m$$

Using a simple prior, $\alpha_j \equiv 1$ for all j, we can compute and plot the model scores as a function of d; see the figure at right. The (unpenalized) maximum likelihood score (dotted line) continues to increase with d; note that its slightly jagged appearance is due to differences in how the bin edges align with the observed data values as d increases. The marginal likelihood and BIC scores, on the other hand, increase to about $d = 3$ bins, then decrease, suggesting that we probably do not have enough data to support estimating a more finely-grained histogram than $d = 3$.



2.4 Convexity

A convex function is one for which, for any pair of points x and x',

$$f(\alpha x + (1-\alpha)x') \leq \alpha f(x) + (1-\alpha) f(x') \qquad \alpha \in [0,1] \tag{2.6}$$

i.e., the line segment connecting $f(x)$ and $f(x')$ lies above the function $f(\cdot)$. A function is called *strictly* convex if the inequality holds strictly.

Convexity can also be framed in terms of the derivatives of f. Two equivalent definitions to (2.6) are:

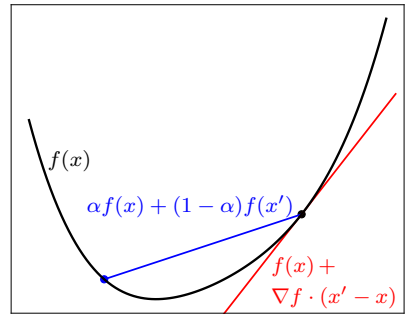
1. f is convex if and only if for all x and x',

$$f(x') \geq f(x) + \nabla f(x) \cdot (x' - x)$$

   where $\nabla f(x)$ is the gradient of f at x (or any subgradient).

2. f is convex if and only if for all x, $\nabla^2 f(x)$ is positive semi-definite.

Convex functions are extremely useful in optimization and machine learning. It is easy to see, for example, that if a function is convex, any local minimum must also be a global minimum. If the function is strictly convex, the global minimum is unique; if not, the minima will form a convex set[6]. Convex functions are thus relatively easy to optimize efficiently.



In the context of random variables, convex functions obey **Jensen's inequality**,

$$\mathbb{E}[f(x)] \leq f(\mathbb{E}[x])$$

which is itself useful in a wide range of contexts.

A *convex optimization problem* is a minimization of a convex function $f(x)$, over a convex constraint set $x \in \mathcal{S}$. In general, optimization of a convex function over a convex region is nicely behaved – $f(x)$ cannot have any non-global optima, and we cannot become locally "trapped" by the constraint set, so local optimization methods such as gradient descent can be guaranteed to converge to the global optimum.

Example 2-23 : Convexity and the exponential family

The negative log-likelihood function of a canonical exponential family distribution is convex in its natural parameters θ; recalling that

$$-\mathcal{L} = m \log Z(\theta) - \sum_i \sum_j \theta_j \phi_j(x^{(i)}) - \sum_i \log h(x^{(i)});$$

we can verify convexity by looking at the matrix of second derivatives, $\nabla^2 \mathcal{L}$. The third term, $\sum h(x^{(i)})$, is a constant with respect to the model parameters θ, while the second term is linear in θ (has second derivative zero). The $\log Z$ term yields,

$$\frac{\partial^2 \log Z(\theta)}{\partial \theta_k \partial \theta_l} = \frac{1}{Z(\theta)} \int \exp\Big(\sum_j \theta_j \phi_j(x)\Big) \phi_k(x) \phi_l(x)\, h(x) \quad = \mathbb{E}[\phi_k(x)\phi_l(x)]$$

so that $\nabla^2 \log Z$ is a covariance matrix of the features ϕ, and thus positive semi-definite.

For the canonical representation of the Bernoulli model $p(X = 1; \eta)$, we saw that the MLE was given by the unique solution $\hat{\eta}_{\text{MLE}} = \log(m_1/m_0)$. If we instead use the "over-parameterized" Bernoulli distribution, $p(X = 1; \eta_0, \eta_1)$, we can see that the set of optima is a convex set, corresponding to the linear constraint $\eta_1 - \eta_0 = \hat{\eta}_{\text{MLE}}$.

[6] A convex *set* is a set in which any convex combination of elements x, x' that are members of the set, i.e., $\alpha\, x + (1-\alpha)\, x'$ for $\alpha \in [0, 1]$, is also a member of the set.

2.5 Information Theory

The field of information theory was first developed to study the fundamental limits of data compression and communications; however, it is deeply rooted in probability and statistics, and many of the concepts that arise in information theory are very useful in reasoning about and characterizing uncertainty. In this section we describe some key concepts in information theory that will be helpful in the sequel. For readers interested in a more in-depth treatment of information theory, see for example Cover and Thomas (2006).

Entropy and Information Theory

Entropy is a measure of our uncertainty about random variables. Suppose that a discrete random variable X has probability distribution $p(X)$. The **Shannon entropy** of X is defined as

$$\mathrm{H}[p(X)] = -\sum_x p(x) \log p(x)$$

where we define $0 \log 0 = 0$ for continuity. We often abbreviate $\mathrm{H}[p(X)]$ as $\mathrm{H}[p]$ or simply $\mathrm{H}[X]$ when the random variable or its distribution are clear from context. If the logarithm is taken base 2, the units of H are called bits; for natural log, they are called "nats". Entropy can also be defined for continuous variables, in which case the summation is replaced by an integral over x.

Example 2-24 : Entropy

Let X_1 be a fair coin flip, i.e., Bernoulli with probability $\frac{1}{2}$. Then the entropy of X_1 is

$$\mathrm{H}[X_1] = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 1 \text{ bit}$$

For comparison, let X_2 be a fair die roll, so that $p(X_2 = x) = \frac{1}{6}$ for all x. Then,

$$\mathrm{H}[X_2] = -6 \cdot \frac{1}{6} \log_2 \left(\frac{1}{6}\right) \approx 2.58 \text{ bits}$$

This matches our intuition; we have more uncertainty about the outcome of a die roll (with 6 equally likely possibilities) than a coin flip (with only two).

Entropy is centrally important to *data compression*, in which we try to represent a sequence of measurements (such as the characters in a text document) using as few bits as possible. Intuitively, entropy corresponds to the "long run" bits per symbol required to transmit the information.

Example 2-25 : Lottery

Suppose we enter the lottery every day. Let X_i be the binary variable indicating whether we won on day i; we would like to communicate our year's results, the sequence $[X_1, \ldots, X_{365}]$, as compactly as possible. We'll assume that our odds of winning stay constant, so that X_i is i.i.d. and equals 1 with some small probabability p.

Obviously, we could communicate our results using 365 bits, just by transmitting the sequence of wins and losses. But, our sequence is not very random – it most likely consists of all zeros (no lottery wins). We can compress our sequence using a variable length code: for example, we could just send the one-bit message "0" if we lost every day; if we won once, we send "1" followed by which day we won (in binary), followed by "0"; and so on. With very high probability, we will not need to send more than one bit of information! The fact that the X_i are relatively non-random (have low entropy) can be exploited to construct a compressed representation.

Entropy is maximized by a uniform distribution over the values of X, with $\mathrm{H} = \log d$. It is minimized by a deterministic distribution (all mass on one value), in which case $\mathrm{H} = 0$.

Kullback-Leibler Divergence

The Kullback-Leibler divergence (sometimes called KL-divergence or relative entropy) measures the difference between two distributions $p(X)$ and $q(X)$. Defined by

$$D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

it has the property that $D(p||q) \geq 0$ for all distributions p, q, with equality if and only if p and q are equal.[7]

Note that the KL-divergence is not a true distance metric: it is not symmetric, i.e., $D(p||q) \neq D(q||p)$ in general, and it also does not obey a triangle inequality. In some sense, it measures q's ability to "cover" the probability mass of p; if $q(x)$ is small at any point x where $p(x)$ is not, the divergence $D(p\|q)$ will be high.

In the context of learning, the log-likelihood of a data set D can be framed in terms of a KL-divergence:

$$\mathcal{L}(q) = -D(\hat{p}(x\,;\,D)\|q) + \mathrm{H}[\hat{p}(x\,;\,D)]$$

where $\hat{p}$ is the empirical distribution of the dataset D. Since $\mathrm{H}[\hat{p}]$ does not depend on the model q, minimizing the divergence $D(\hat{p}\|q)$ is equivalent to maximizing the log-likelihood of the model q. In this sense, the maximum likelihood parameters give us the "nearest" distribution in our family to the empirical distribution $\hat{p}$. In *information geometry*, the minimizing distribution q is called a *moment projection* or *m-projection* of $\hat{p}$; for more details see, e.g., Amari (2016).

Mutual Information

Mutual information measures the amount of information that observing one random variable tells us about another; it quantifies their degree of (in)dependence. The mutual information between two variables X and Y is given by

$$\begin{aligned}\mathrm{I}[X,Y] &= \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \\ &= D(\,p(X,Y)\,||\,p(X)\,p(Y)\,)\end{aligned}$$

[7] For continuous random variables, equality holds up to a set of measure zero.

i.e., the KL-divergence between the joint distribution $p(X, Y)$ and an independent distribution defined by the product of their marginal probabilities. By its KL-divergence form, it is clear that I is non-negative, and zero if and only if X and Y are independent.

Rearranging, we can also show that

$$\mathrm{I}[X, Y] = \mathrm{H}[X] + \mathrm{H}[Y] - \mathrm{H}[X, Y]$$

where $\mathrm{H}[X, Y]$ is the joint entropy, or entropy of the joint random variable (X, Y).

Conditional Entropy

We define the conditional entropy of X given Y, written $\mathrm{H}[X|Y]$, to be the expected entropy of a variable *after* measuring Y, averaged over the possible values of Y:

$$\begin{aligned}
\mathrm{H}[X|Y] &= -\sum_y p(Y = y) \sum_x p(X = x|Y = y) \log p(X = x|Y = y) \\
&= -\sum_{x,y} p(x, y) \log p(x|y) \\
&= -\sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(y)} = \mathrm{H}[X, Y] - \mathrm{H}[Y]
\end{aligned}$$

Comparing this to the mutual information, we see that $\mathrm{I}[X, Y] = \mathrm{H}[X] - \mathrm{H}[X|Y]$; since I is non-negative, we must have $\mathrm{H}[X] \geq \mathrm{H}[X|Y]$, i.e., conditioning on observations can only reduce our uncertainty about X on average. Note that this holds only on average over Y, not for specific values: for any particular value y, the entropy $\mathrm{H}[X|Y = y]$ of $p(X|Y = y)$ may be larger or smaller than $\mathrm{H}[X]$.

Example 2-26 : Information and Conditional Entropy

Suppose we decide to build a model of our commuting habits ($C \in \{\text{walk}, \text{bike}, \text{drive}\}$) in different weather ($R \in \{\text{clear}, \text{rain}\}$). In southern California, it rains on about 10% of days; on clear days, we walk or bike to work, while on rainy days we walk (with an umbrella) or drive, with probabilities:

$$p(R = \text{rain}) = 0.1$$

$$\begin{aligned}
P(C = \text{walk}|R = \text{clear}) &= 0.9 \quad P(C = \text{bike}|R = \text{clear}) = 0.1 \quad P(C = \text{drive}|R = \text{clear}) = 0.0 \\
P(C = \text{walk}|R = \text{rain}) &= 0.5 \quad P(C = \text{bike}|R = \text{rain}) = 0.0 \quad P(C = \text{drive}|R = \text{rain}) = 0.5
\end{aligned}$$

How much information does observing the weather give about our commute?

We first calculate the marginal probability $p(C)$ and its entropy, $\mathrm{H}[C]$:

$$p(C = \text{walk}) = 0.86 \qquad p(C = \text{bike}) = 0.09 \qquad p(C = \text{drive}) = 0.05$$

$$\Rightarrow \qquad \mathrm{H}[C] = -0.86 \log_2(0.86) - 0.09 \log_2(0.09) - 0.05 \log_2(0.05) \quad \approx 0.72 \text{ bits}$$

Then, we can calculate the conditional entropy:

$$\begin{aligned}
\mathrm{H}[C|R = \text{clear}] &= -0.9 \log_2(0.9) - 0.1 \log_2(0.1) - 0 \log_2(0) \quad \approx 0.47 \text{ bits} \\
\mathrm{H}[C|R = \text{rain}] &= -0.5 \log_2(0.5) - 0 \log_2(0) - 0.5 \log_2(0.5) \quad = 1.0 \text{ bits}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \qquad \mathrm{H}[C|R] &= p(R = \text{rain})\mathrm{H}[C|R = \text{rain}] + p(R = \text{clear})\mathrm{H}[C|R = \text{clear}] \\
&\approx (0.9)(0.47) + (0.1)(1.0) \quad \approx 0.52 \text{ bits}
\end{aligned}$$

giving the mutual information,

$$\mathrm{I}[R, C] = \mathrm{H}[C] - \mathrm{H}[C|R] \approx 0.72 - 0.52 = 0.2 \text{ bits}$$

We can see that, in this example, observing $R =$ rain actually makes C more uncertain, i.e., $\mathrm{H}[C] < \mathrm{H}[C|R = \text{rain}]$, but on average, observing R makes C less uncertain, so that $\mathrm{H}[C] > \mathrm{H}[C|R]$; the mutual information is always nonnegative.

2.6 Change-of-Variable Models

The classical probability distributions we have considered so far, such as the Gaussian distribution, are relatively simple and well-behaved. Unfortunately, many real-world data sets may not fit these classic forms. One simple but powerful technique we can use to extend our models is to use a change of variables, in which our variable of interest, X, is defined in terms of an invertible transformation $f(\cdot)$ of some other variable, Z.

A classic example of such a model is the **lognormal** distribution. We say that a random variable X is lognormally distributed if its log, $Z = \log(X)$, has a normal (Gaussian) distribution. To draw a sample of a lognormal X is easy: we can draw a sample from the base distribution, $z \sim p(Z)$, and then transform z to get the value $x = \exp(z)$.

Given this definition, what is the probability density function of X? We can answer this question by defining our transformations, $f(Z) = \exp(Z)$ and $g(X) = f^{-1}(X) = \log(X)$, and writing the probability,

$$\Pr[a \leq X \leq x] = \int_a^x p_X(X)dX = \Pr[g(a) \leq Z \leq g(x)] = \int_{g(a)}^{g(x)} p_Z(Z)dZ$$

i.e., the probability that X is in the interval $[a, x]$ is the same as the probability that Z is in the interval $[g(a), g(x)]$[8]. Now, the density $p(x) = \frac{d}{dx} \Pr[-\infty \leq X \leq x]$, so we have by the chain rule that

$$p(x) = \frac{d}{dx} P_Z(\, g(x)\,) = p_Z(\, g(x)\,)\, g'(x).$$

where $P_Z(z) = \int_{-\infty}^{z} p_Z(z)dz$ is the cumulative distribution function of Z. More generally, for any invertible transformation $Z = g(X)$, $p_X(x) = p_Z(\, g(x)\,)\, |g'(x)|$.

Intuitively, this result tells us that, if we wish to know the probability that X is very close to x, i.e., $x \pm \epsilon$, we need to know how quickly Z changes over that interval, i.e., the probability that Z is in $z \pm \epsilon\, g'(x)$. For the lognormal distribution, this means

$$p(x) = \Big(\frac{1}{2\pi\sigma^2}\Big)^{\frac{1}{2}} \exp\Big(-\frac{1}{2}\big(\log(x) - \mu\big)^2/\sigma^2\Big)\, \frac{1}{x}$$

since for $g(x) = \log(x)$ the derivative $g'(x) = \frac{1}{x}$.

The same trick extends naturally to multivariate distributions, in which $z = g(x)$ is a vector-valued function, $z = [z_1, \ldots z_n]$ and $z_i = g_i(x)$. In this setting, the change of variables gives us

$$p_X(x) = p_Z(\, g(X)\,)\, |\det \nabla_X g|,$$

where $\nabla_X g$ is the Jacobian matrix of the transformation $g(\cdot)$, i.e., $(\nabla_X g)_{ij} = \frac{\partial g_i(x)}{\partial x_j}$, $|\cdot|$ is the absolute value, and $\det A$ indicates the determinant of a matrix A.

[8] Here we use the fact that $g(x)$ is strictly increasing; if $g(x)$ is instead decreasing, the interval is $[g(x), g(a)]$ instead.

Copula Models

Copula models are a group of models popular in econometrics and finance that aim to separate the modeling of each individual variable's marginal distribution from the modeling of the interactions among the variables. They separate these two aspects by writing the joint cumulative distribution in terms of the individual variables' cumulative distributions and an interaction function C, called a **copula**. For example, for $X = [X_1, X_2]$ we can write,

$$\Pr[X_1 \leq x_1, X_2 \leq x_2] = C(\, P_1(x_1), P_2(x_2)\,) \quad P_1(x_1) = \Pr[X_1 \leq x_1] \quad P_2(x_2) = \Pr[X_2 \leq x_2]$$

where P_1, P_2 are the cumulative distributions of X_1 and X_2, respectively, and C is the copula function, a cumulative distribution defined on inputs that are uniformly distributed on $[0, 1]$. (If P_i is the cumulative distribution of X_i, then $P_i(X_i)$ is uniformly distributed.)

Viewing this as a change of variables, we can see that

$$p(X_1, X_2) = c(\, P_1(x_1), P_2(x_2)\,)p_2(X_1)p_1(X_2)$$

where c is a copula density (i.e., a density with uniform marginals, specifically the derivative of C). Frequently, the copula function C and density c are themselves defined in terms of transformations of well-known distributions; for example, the Gaussian copula is,

$$c(z_1, z_2) = \mathcal{N}([\Phi^{-1}(z_1), \Phi^{-1}(z_2)]\ ;\ \mu, \Sigma)\ \frac{\partial \Phi^{-1}}{\partial z}(z_1)\ \frac{\partial \Phi^{-1}}{\partial z}(z_2)$$

where Φ is the unit Gaussian CDF. Thus, the copula $C(\cdot)$ and its density $c(\cdot)$ are defined by a change of variables from z_i to $\zeta_i = \Phi^{-1}(z_i)$, where ζ_i is Gaussian distributed.

In general, copula methods allow us to use univariate models for the $p(X_i)$ that are complex and could lead to overfitting if used for the joint distribution. For example, a univariate histogram model might be reasonably accurate given a few hundred data samples, when a multivariate histogram would not. The so-called "curse of dimensionality" in machine learning tells us that such models will need exponentially more data to maintain their estimate quality, since for example, gridding the n-dimensional unit hypercube $[0, 1]^n$ to some resolution ϵ would require ϵ^{-n} bins. In contrast, copula models allow complex univariate models to be combined with simple models of the variables' interaction (e.g., through a Gaussian covariance Σ).

While the maximum likelihood estimator of a copula model requires jointly optimizing the parameters of the copula density $c(\cdot)$ and the marginals $p_i(X_i)$, a common approximate optimization method is to first estimate the p_i, then optimize $c(\cdot)$ with the marginals held fixed. For example, one can define a *semiparametric* model consisting of nonparametric[9] estimates of the P_i (for example, using the empirical cumulative distribution); then holding the P_i fixed, we can optimize the parameters of $c(\cdot)$.

Example 2-27 : Copula Transforms

Consider a data set of high-energy physics measurements, used in the 2004 KDD Cup competition (Caruana et al., 2004). To enable visualization, we take just $m = 1500$ samples and two features X_1, X_2[10] in Figure 2.5(a). We can accurately estimate the marginal densities of X_1 and X_2 (via histograms) and their corresponding cumulative distributions P_1, P_2 from these data. The data are very non-Gaussian; they are strictly positive, and concentrated near the origin. Transforming by $Z_i = P_i(X_i)$ places the data in the unit square, $[0, 1]^2$, with uniform marginal densities, visualized in Figure 2.5(b). Finally, transforming by $\zeta = \Phi^{-1}(Z)$ gives unit Gaussian marginals, as in Figure 2.5(c).

[9] Nonparametric statistical models are models in which the number of parameters grows with the number of data points, for example kernel density estimates (Parzen, 1962) and other models that store and use the data themselves to compute probabilities.

[10] Originally features 20 and 24 in the competition.

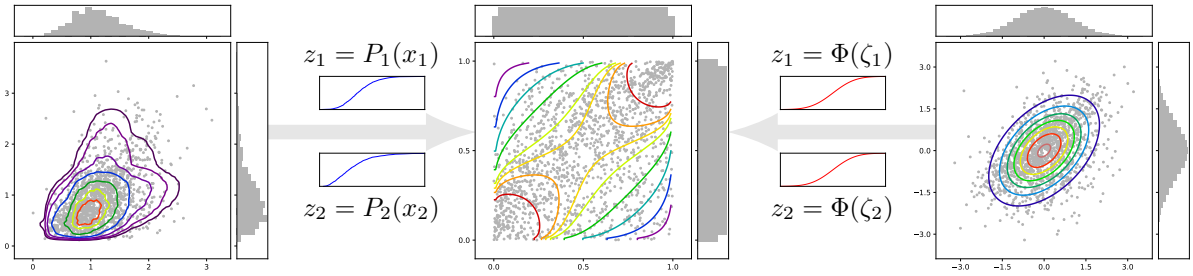


Figure 2.5: Copula transforms on 2004 KDD Cup data. (a) Scatter plot of the data; estimating the univariate CDFs P_1, P_2 allows us to transform into (b), where the marginals of $[z_1, z_2]$ are uniform. (c) Transforming by Φ^{-1} gives $[\zeta_1, \zeta_2]$ with unit Gaussian marginals. Finally, a Gaussian density $\mathcal{N}(\zeta)$ can be transformed forward to find its induced densities on Z and X, shown using contours.

Then, a correlation coefficient defines a Gaussian density $\mathcal{N}(\zeta)$ with unit marginal variances, and the transformations $Z = \Phi(\zeta)$ and $X_i = P(Z_i)$ induce densities in the copula space Z and original features X, whose contours are displayed in each panel, and we can optimize the likelihood of the data $\{x^{(s)}\}$ over the correlation coefficient to select the value of ≈ 0.73, displayed. The positive correlation in ζ induces positive correlations in Z and X, but with very different marginal densities.

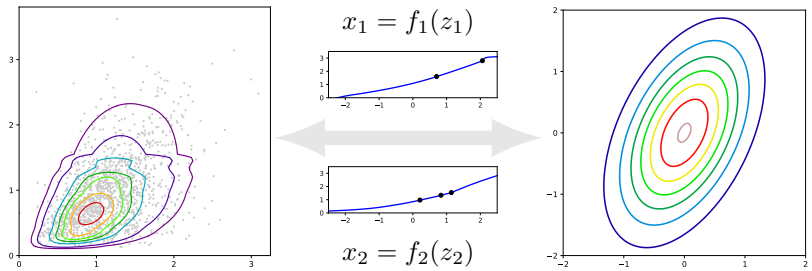
Normalizing Flow Models

Normalizing flow models define a parametric form for the change-of-variables transformation f which is, by construction, invertible: $X = f(Z) \Leftrightarrow Z = g(X) = f^{-1}(X)$. Combined with a simple base distribution $p(Z)$, this gives a flexible class of probability distributions on x.

For example, a Gaussian copula model is a special type of normalizing flow, in which the base distribution $p(Z)$ is Gaussian, and the transformation $f(Z)$ is defined by the desired cumulative distribution functions of the individual dimensions. If we select $p(Z)$ to capture the co-dependence between the dimensions, and the transforms $f_i(z)$ to operate independently on each feature, $f_i(z) = f_i(z_i)$, we can obtain a copula-like model parameterized by our functions $\{f_i\}$:

Example 2-28 : Copula-like normalizing flow

Let us select a Gaussian base distribution $p(Z)$, parameterized by its covariance, and define univariate transformations $f_i(Z_i)$ using **splines**: piecewise low-order polynomial functions defined by the positions of several *knots* with, for example, quadratic functions interpolating between the knots. Then, we can use autodifferentiation to jointly optimize over $p(Z)$ and the f_i to maximize the induced likelihood of our data, $\{x^{(s)}\}$. This model effectively uses the f_i as a parameterized stand-in for the cumulative distribution transformations $P_i^{-1}(\Phi(\zeta))$, rather than explicitly parameterizing the P_i, while leaving the dependence between the X_i defined by the parameters (covariance) of $p(Z)$.



More generally, however, we can define multivariate transformations to $p(Z)$ that induce dependence among the dimensions of X. Consider a *fixed* base distribution $p(Z)$, for example an independent unit Gaussian, followed by a parameterized multivariate transformation $X = f(Z)$. Now $p(Z)$ has no parameters to optimize, and the dependence between the X_i are induced by the function $f(\cdot)$.

The difficulty with this type of model, particularly for higher-dimensional X, lies in computing the Jacobian determinant of the transformation f. However, a useful trick is to define f as a composition of a sequence of transformations, $f(Z) = f_T(\, f_{T-1}(\, \ldots\ f_1(Z)\,)\,)$. This gives an overall change of variable transformation with,

$$\log p_X(X) = \log p_Z(f^{-1}(X)) - \sum_t \log |\nabla f_t|,$$

so that if each f_t's Jacobian determinant is easy to compute, so is the full transformation.

The building blocks f_t should provide for a nonlinear transformation that correlates the X_i, while ensuring an easy-to-evaluate Jacobian. To provide this, we can use a *coupling transform* (Dinh et al., 2015), which transforms some dimensions ("unmasked" dimensions) in ways that depend on the others ("masked" dimensions). As a concrete example, take the conditional affine transformation

$$Z' = [Z'_1, Z'_2] \;=\; f_1(Z_1, Z_2) = [\; Z_1\;,\;\; \alpha_1(Z_1)\, Z_2 + \beta_1(Z_1)\;], \tag{2.7}$$

where $\alpha_1(\cdot)$ and $\beta_1(\cdot)$ are potentially complex functions of Z_1, typically defined by a small multi-layer neural network. The function f_1 is easily inverted, since $Z'_1 = Z_1$, allowing us to calculate the values of α_1, β_1 and then find $Z_2 = (Z'_2 - \beta_1)/\alpha_1$. Moreover, the Jacobian's determinant is easy to calculate, since the Jacobian is lower-triangular (Z'_2 depends on Z_1, but Z'_1 does not depend on Z_2), and thus the determinant is the product of the diagonal entries – i.e., in our example $\log |\nabla f_1| = \log |\alpha_1(Z_1)|$.

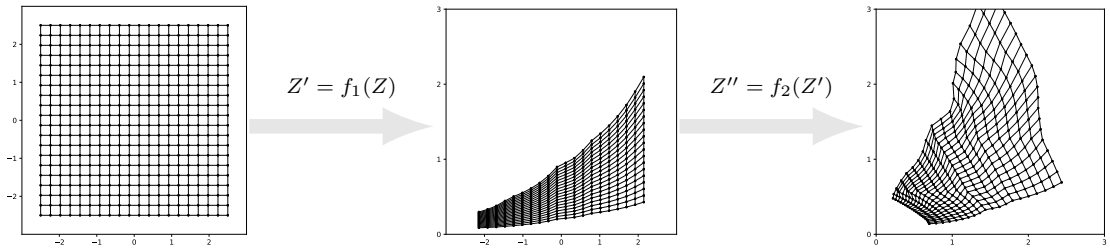
Obviously, f_1 does not transform Z very significantly (Z_1 is not altered at all!), but in subsequent functions we can change which variables are masked, e.g.,

$$Z'' \;=\; f_2(Z'_1, Z'_2) = [\; \alpha_2(Z'_2)\, Z'_1 + \beta_2(Z'_2)\;,\;\; Z'_2\;]; \tag{2.8}$$

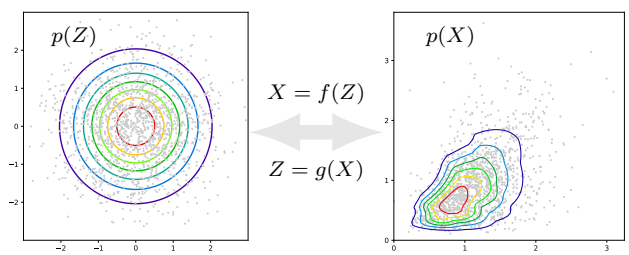
then, by cascading many simple transformations $f_1 \ldots f_T$ we obtain a complex overall transformation f.

Example 2-29 : Conditional affine normalizing flows

Consider applying a sequence of two conditional affine normalizing flows, so that $X = Z''$ as defined in Eq. (2.7)–(2.8). We train the parameters defining $\{\alpha_1(Z_1), \beta_1(Z_1), \alpha_2(Z'_2), \beta_2(Z'_2)\}$ to optimize $p(X)$; we can visualize the resulting transformations by applying them to a collection of equal-area patches:



We see that although f_1 leaves Z_1 (horizontal axis) unaltered, the affine transformation's dependence on the value of Z_1 allows it to shift and expand the range of Z_2' more on the right (positive Z_1) than on the left (negative Z_1). Then, the transformation f_2 leaves Z_2' unchanged, but shifts and scales Z_1' by an amount that depends on Z_2', giving a more complex overall transformation than either f_1 or f_2 alone. The determinant of ∇f measures the expansion or contraction of the equal-area regions of Z into differently-sized regions in X, and its effect on their associated probability mass.



Since the determinants of each f_t are easy to compute, the distribution $p(X)$ induced by our unit normal base distribution $p(Z)$ followed by $f(Z) = f_2(\, f_1(Z)\,)$ remains easy to calculate, while the transforms enable $p(X)$ to have both dependence between X_1, X_2 and non-Gaussian shapes. To capture more complex dependencies, the function $f(\cdot)$ can be made increasingly flexible by adding more transformations f_t to the sequence.

There are a number of useful variations on this general theme of coupling transformations. For example, we can define the transformation of each Z_i in terms of all the preceding dimensions, $[Z_1 \ldots, Z_{i-1}]$, termed an "autoregressive" transform; this retains the lower-triangular form of the Jacobian (Huang et al., 2018) but can be slower to invert than a block coupling transform in higher dimensions. Another route is to apply more sophisticated transforms to Z_i than a simple affine update. For example, neural spline flows (Durkan et al., 2019) transform Z_i using a montontic spline function, akin to the spline transforms used in Example 2-28. Like the conditional affine transform, we can divide the dimensions of Z into masked and unmasked sets, then apply a basic spline transform to the masked dimensions and a conditional spline transform to the unmasked dimensions, in which the parameters of the spline – typically the knot locations and derivatives of the spline at those points – are defined by the output of a neural network given the value of the masked dimensions. By using more flexible transformations such as splines, we may be able to transform Z sufficiently using far fewer coupling transforms than would be required using only simpler, conditional affine transforms.